Experiment 3: Transistor Biasing

Pre-Laboratory

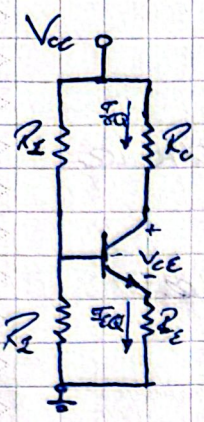


Fig. 1a

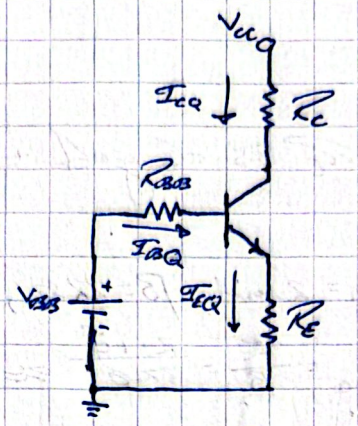
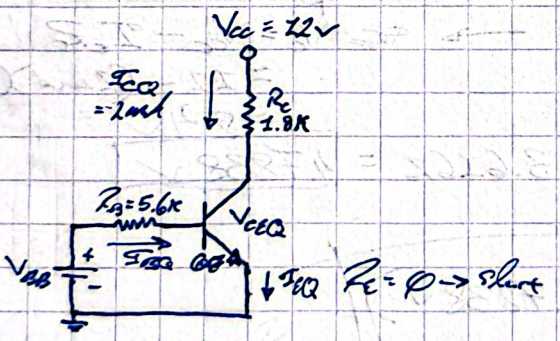


Fig. 1b

Assume $V_{cc} = 12V$

A) Consider the circuit in Fig. 1.b with $R_C = 1.8k\Omega$, $R_B = 5.6k\Omega$, and $R_E = 0\Omega$

B) Calculate V_{BB} so that $I_C = 2mA$. Assume $\beta = 220$ and $V_{BE} = 0.7V$. Find the Q-point.



$\beta = 220$

$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{2mA}{220} = 0.009\,mA$

$= 0.01\,mA$

$= 9\mu A$

$V_{BE} = V_B - V_E = 0.7$

$V_E = 0$ (ground) $\rightarrow V_B = 0.7$

$I_{BQ} = \frac{V_{BB} - V_B}{R_B} \longrightarrow 9\,\mu A = \frac{V_{BB} - 0.7}{5.6K}$

Q-point: (2mA, 8.4V)

$0.00504 = V_{BB} - 0.7 \longrightarrow V_{BB} = 0.756\,V$

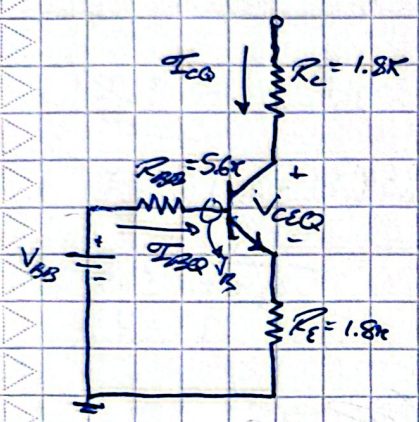
$\boxed{0.75004\,V}$

$V_{CEQ} \rightarrow V_C = V_{CC} - I_C R_C \longrightarrow I_C = \frac{V_{CC} - V_C}{R_C}$

$V_C = 12 - (2mA)(1.8) = \boxed{8.4V} = V_{CEQ}$

c) If β changes to 150, what is the new I_C from your circuit simulation? (Consult with your lab instructor how to change the current gain β for the 2N2222 model parameter in multisim)

D) Repeat (B) above if R_E is changed to 1.8k Ω



$I_{CQ} = 2mA$, $\beta = 220$, $V_{BE} = 0.7$

$I_{BQ} = \frac{I_C}{\beta} = \frac{2 \cdot 10^{-3}}{220} \approx 9\mu A$

$I_E = I_C + I_B = 2mA + 9\mu A = 2.009 mA$

$V_E = I_E R_E = (2.009 mA)(1.8K) = 3.6162 V$

$V_B = V_E + V_{BE} = 3.6162 + 0.7 = 4.312 V$

$I_B = \frac{V_{BB} - V_B}{R_B} \longrightarrow V_{BB} = I_B R_B + V_B$

$= (9\mu A)(5.6K) + 4.312$

$V_{BB} = 4.362 V$

$I_C = \frac{V_{CC} - V_C}{R_C} \longrightarrow V_C = V_{CC} - I_C R_C$

$= 12 - (2mA)(1.8K)$

$= 8.4 V$

$V_{CEQ} = V_C - V_E = 8.4 - 3.6162 = \boxed{4.7838 V}$

Q-point : (2mA, 4.7838 V)

E) Consider the circuit for Fig. 3 with $R_C = R_E = 1.8k\Omega$. Calculate the values for R_1, R_2 so that $I_{CQ} = 2mA$. Use a sinusoidal input (small signal gain to give voltage of 20mV) and current gain of 150 in your circuit simulation. Plot V_i, V_C, V_E, and V_{CE} versus time

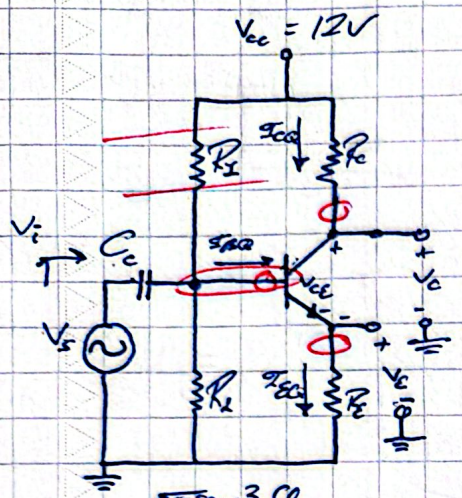


Fig 3.0

$R_C = R_E = 1.8k$

$I_{CQ} = 2mA$

$V_i = V_S = 20mV_{p-p}$

$A = 150 = \beta$

$V_{cc} = 12$

$V_{CE} = V_C - V_E$

$= 8.4 - 3.84$

$= \boxed{4.56V = V_{CE}}$

$I_{BQ} = \frac{I_C}{\beta} = \frac{2mA}{150} = 0.013mA$

$I_E = I_C + I_B = 2mA + 0.013mA =$

$= 2.013mA$

$V_E = I_E R_E = (2.013mA)(1.8k)$

$V_B = 8.4 - 0.7 = 7.7V$

$V_B - V_C = 0.7$

$V_i = 10V_p$

$I_B = \frac{I_C}{\beta}$

$2mA = \frac{12 - V_C}{1.8k}$

$3.6 = 12 - V_C$

$\boxed{V_C = 8.4V}$

$V_{th} = \frac{R_2}{R_1 + R_2} \cdot 12$

$I_{BQ} = 0.013mA = \frac{(V_{th}) - V_B}{\left(\frac{R_1 R_2}{R_1 + R_2}\right)} = \frac{\left(\frac{R_2}{R_1 + R_2} \cdot 12\right) - V_B}{\left(\frac{R_1 R_2}{R_1 + R_2}\right)}$

$(0.013mA)\frac{R_1 R_2}{R_1 + R_2} = \left(\frac{R_2}{R_1 + R_2} \cdot 12\right) - V_B$

$(0.013mA) R_1 R_2 = 12 R_2 - V_B(R_1 + R_2)$

If $R_1 = 1k \longrightarrow (0.013mA)(1k) R_2 = 12 R_2 - 7.7(1k + R_2)$

must assume one resistor value

$0 = 12 R_2 - 7.7(1k + R_2) - 0.013 R_2$

$R_2 \approx 564\Omega$

$V_E = I_E R_E = (I_C + I_B) R_E = \left(I_C + \frac{I_C}{\beta}\right) R_E = \left(2mA + \frac{2mA}{150}\right)(1.8k)$

$V_E = 3.84$